INTERNATIONAL DEVELOPMENT ASSOCIATION AND
INTERNATIONAL MONETARY FUND

BURKINA FASO

Joint World Bank/IMF Debt Sustainability Analysis

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The results of the Debt Sustainability Analysis (DSA) are similar to those of the previous DSA. Burkina Faso's risk of debt distress is high, because the present value (PV) of debt-to-exports ratio is projected to breach its policy-dependent indicative threshold under the baseline scenario and under the stress tests.[1] *However, the breach is projected to occur about 10 years later than in the previous DSA, reflecting better exports prospects over the period. Other debt burden indicators remain below their policy-dependent indicative thresholds under the baseline and stress tests scenarios. In the context of the new PRSP adopted in late 2010, Burkina Faso plans to scale up investment, particularly in infrastructure, to accelerate growth and reduce poverty. The DSA results indicate that financing needs for higher public investment would have to be met mostly with grants and concessional loans to support long-term debt sustainability.*

I. INTRODUCTION

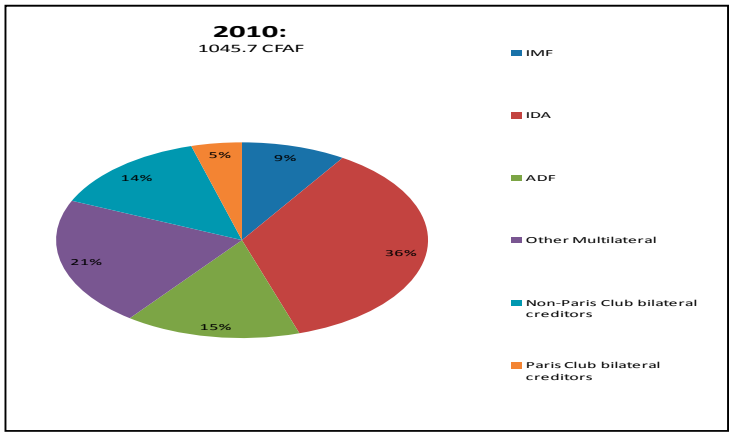
1. **Burkina Faso's stock of debt comprises public and publicly guaranteed external debt and domestic debt.** At end-2010, external debt amounted to 23.9 percent of GDP with some 80 percent owed to multilateral creditors (Figure 1) and about a third

[1] Under the LIC DSF, it is expected that better policies and stronger institutions help countries manage higher levels of debt. Hence, the thresholds for the DSA are policy-dependent and determined based on the World Bank's Country Policy and Institutions Assessment (CPIA) classification. Based on the average CPIA score in 2008–10, Burkina Faso is ranked as a "medium performer". The relevant indicative thresholds for countries in this category are: 40 percent for the present value (PV) of debt-to-GDP ratio; 150 percent for the PV of debt-to-exports ratio; 250 percent for the PV of debt-to-revenue ratio; 20 percent for the PV of debt service-to-exports ratio; and 30 percent for the PV of debt service-to-revenue ratio.

denominated in Euro, which mitigates the impact of exchange rate fluctuations.[2] Government bonds (Text Table 1) dominate the domestic debt.

2. **External debt policy has remained prudent. Burkina Faso has systematically sought grants and concessional loans to cover financing needs**. Hence, under the Fund-supported program, the 35 percent grant element requirement for public and publicly-guaranteed external debt is strictly enforced, and investment projects are consistently analyzed by a National Public Debt Committee before loan negotiations are initiated. The DSA assumes that during 2017–31, the structure of Burkina Faso's external debt would not change (see Figure 1), and therefore, the average grant element would remains above 35 percent.

Text Figure 1. Burkina Faso—External Debt Composition by Creditor, 2010



[2] Burkina Faso's currency, the CFA Franc which is common to 6 other members of the West African Economic and Monetary Union, is pegged to the Euro at a fixed exchange rate.

Text Table 1. Burkina Faso: Stock of Public Debt, 2006–2010

	2006	2007	2008	2009	2010
			(In billions of CFAF)		
Total debt stock	**687.1**	**713.1**	**883.5**	**1,029.6**	**1,185.2**
External	603.8	601.6	786.4	867.7	1,045.7
Multilaterals	440.1	473.0	604.7	688.5	853.1
Bilaterals	163.8	128.6	181.8	179.2	192.6
Domestic	83.3	111.6	97.1	162.0	139.5
			(In percent of GDP)		
Total debt stock	**21.7**	**21.9**	**23.9**	**26.1**	**27.1**
External	**19.0**	**18.5**	**21.3**	**22.0**	**23.9**
Multilaterals	13.9	14.5	16.4	17.5	19.5
Bilaterals	5.2	4.0	4.9	4.5	4.4
Domestic	**2.6**	**3.4**	**2.6**	**4.1**	**3.2**

Sources: IMF staff and Burkinabè authorities.

3. **This DSA is carried out with a macroeconomic outlook reflecting the expected improvement in economic prospects over the medium to long term** (Box 1, Table 1). The outlook underpinning the DSA accounts for the impact of the projected increase in public investment, particularly in infrastructure to achieve growth and poverty reduction objectives outlined in the authorities' new PRSP adopted in December 2010. The outlook reflects also the projected expansion in the mining sector, through higher gold production and the development of mines for other minerals, better prospects in the cotton sector, and increased diversification in the agriculture sector. Consequently, real GDP growth is projected to average 7.3 percent in 2017–31, compared with 6.2 percent in the previous DSA. The noninterest current account deficit, although remaining moderate, is forecast to be higher than in the 2010 DSA because demand for oil and capital goods imports is expected to remain strong, in line with the projected increase in public investment, and improved economic prospects. Similarly, the external debt path is less favorable than in the previous DSA, with the rate of debt accumulation increasing steadily during 2011–20, before stabilizing over the long-term. Fiscal consolidation efforts are expected to be intensified to increase the domestic resource contribution to investment financing: tax revenue is forecast to increase from 15.6 percent of GDP in 2017 to 17.3 percent of GDP in 2031 thanks to a combination of increased revenue from mining and private sector activities and efficiency gains in tax administration; and the overall fiscal balance (including grants) is projected to improve more markedly than under the previous DSA. The macroeconomic framework also assumes that foreign aid would be consistent with historical trends in the medium-term and gradually decline as revenue performance strengthens.

Box 1. Burkina Faso: Macroeconomic Assumptions Underlying the DSA

Real GDP is projected to average 6.4 percent in the medium term, supported by the expected increase in gold production and global prices, improved production in the cotton sector, notably thanks to higher productivity with the planned extended use of genetically modified cotton seeds, and the projected buoyant activity in the services sector. Growth projections are higher than in the 2010 DSA, reflecting a more positive outlook: the 2010 DSA outlook, which was marked by uncertainties and downside risks following the 2008–09 terms of trade and climatic shocks that adversely affected economic activity. In the longer term, growth prospects are expected to be enhanced through: (i) higher and more efficient public investment, particularly in infrastructure. Key projects under the PRSP include the Samendeni dam, the Bagre growth pole, and the Donsin airport; (ii) increased energy supply as planned investments in the sector are completed; (iii) implementation of diversification initiatives scheduled in the new PRSP with greater participation from the private sector and; (iv) continued implementation of growth-enhancing structural reform measures. In this vein, an efficient implementation of the financial sector strategy will be critical in broadening access to financial services and supporting private sector development. Real GDP is forecast to expand by 7.4 percent on average during 2017–31, despite a projected deceleration in export growth, as explained below, thanks to the expected impact of higher public investment on infrastructure and public works, and the anticipated growth in the services sector.

Inflation is expected to remain moderate and in the low-single digits, as the authorities plan to intensify measures initiated in recent years to support agriculture production, which have resulted in increased supply of agricultural food products and helped contain inflationary pressures. On this basis, and assuming moderate increase in import prices, as well as continued prudent regional monetary policy, inflation is forecast to remain below 3 percent during the projection period.

The external current account deficit, excluding grants, is forecast at about 10 percent of GDP on average in the medium to long term. Under the baseline scenario, higher export growth assumptions compared with the previous DSA reflect the expected improvement for gold and cotton export earnings as WEO projections show that gold and cotton prices will on average remain at the current high levels for the next two decades. Over the long term, export growth is projected to remain strong, but decelerating towards the end of the period. These projections reflect conservative assumptions on prospective production in the agriculture, and cotton sectors in view of uncertainties on weather conditions over the long term; and a moderate increase in gold export volume taking into account the lifespan of existing mines. Imports are expected to remain strong as investment increases and oil prices are projected to remain high. The external current account (excluding official transfers) is projected to widen from 9.9 percent of GDP in 2012 to about 11 percent of GDP in 2016, before declining to 8 percent in 2027 and 6½ percent of GDP in 2031.

The basic primary fiscal deficit is expected to average 1.4 percent of GDP during 2012–16, and to decline over the long term. This mainly reflects a projected strong revenue mobilization based on increased efficiency in revenue collecting agencies, measures to curb fraud and tax evasion, and to modernize tax administrations. These assumptions are more optimistic than under the previous DSA. They take into account the impact of the 2010 tax reform strategy and prospects for higher taxes in the mining sector. The anticipated higher revenue mobilization would create additional fiscal space for increased government investment; and externally financed investment would average 5.8 percent of GDP in 2017–30, thus strengthening the authorities' efforts to accelerate growth. The DSA macroframework assumes that current transfers would average about 4 percent of GDP during 2017–31, while investment outlays would reach some 13 percent of GDP over the medium to long term, compared with about 11 percent in the last 10 years. External budget support is assumed to average 3.6 percent of GDP during 2017–31

4. **Debt dynamics are positively affected by the higher growth and export assumptions.** DSA results show that under the baseline scenario, at the end of the projection period, the PV of debt-to-GDP ratio deteriorates only marginally compared with the previous DSA, despite a higher rate of external debt accumulation. In the external sector, the PV of the debt-to-exports ratio breaches the policy-dependent threshold nearly 10 years later than in the previous DSA thanks to the robust performance of the export sector, and decelerates thereafter. Nevertheless, those improvements could be jeopardized by terms of trade and weather-related shocks that could affect Burkina Faso's GDP and export growths.

II. External Debt Sustainability Analysis

5. **The external debt sustainability analysis shows that the risk of debt distress is high** because the PV of debt-to-exports ratio deteriorates during the projection period (Table 2 and Figure 1). In the baseline scenario, it is projected to increase from 71 percent in 2011 to 114 percent by 2021, and 176 percent in 2031, well-above the 150 percent threshold. The DSA shows comparable results under alternative scenarios. In particular, the PV of debt-to-exports ratio would reach 273 percent by 2031 under the most extreme shock, which assumes less favorable borrowing conditions for new debt (an increase in interest rate of 2 percentage points). These results suggest that Burkina Faso's long-term debt profile would deteriorate significantly if investment-financing needs were covered with nonconcessional loans. An alternative scenario assuming a lower grant element over the long term produces similar results (Figure 2). The PV of debt-to-exports ratio would also breach the policy-dependent indicative threshold in the case of a shock that would prevent the projected improvement in exports earnings, or other key macroeconomic variables.

6. **The DSA results indicate that other external debt indicators would remain below their policy-dependent thresholds**. Under the baseline and alternative scenarios, the PV of debt-to-GDP ratio and the PV of debt-to-revenue ratio increase somewhat, without breaching their respective indicative thresholds during the projection period. For both indicators, the standard stress tests do not point to particular vulnerability to shocks (Table 2). In particular, under the assumption of less favorable borrowing conditions for new loans, the PV of debt-to-GDP ratio increases from 17 percent in 2011 to about 24 percent in 2031 (with a 40 percent threshold); and the PV of debt-to-revenue ratio increases from 108 percent in 2011 to 123 percent in 2031 (with an indicative threshold of 250 percent).

7. **Results from DSA stress tests are consistent with downside risks to Burkina Faso's medium-term outlook**, which are related to the economy's vulnerability to terms of trade shocks and fluctuations in climatic conditions that may affect economic growth and export earnings. While these shocks may affect exports and growth and increase the PV of debt-to-exports ratio, sensitivity analysis shows that a positive shock leading to an

export growth of about 8 percent during 2017–31 (compared with 5.3 percent in the DSA framework) would lower the risk of debt distress to moderate levels.

III. Public Debt Sustainability Analysis

8. **The DSA results do not change with the inclusion of domestic debt in the analysis** (Table 3). Burkina Faso's stock of domestic debt amounted to about 3.2 percent of GDP at end- 2010. It comprises government bonds (82 percent) and consolidated liabilities to the banking system. Since the stock of domestic debt is low, and projected to decline based on repayment schedules,[3] DSA results on public debt are similar to the above-mentioned results on external debt.

IV. Debt Management Issues

9. **Debt management capacity needs further improvement.** The authorities have taken measures in recent years to enhance debt management capacity. The World Bank mission (March 2011) on Debt Management Performance Assessment (DeMPA), concluded that Burkina Faso had made significant progress on most debt management indicators, compared with the 2008 DeMPA assessment. However, some weaknesses remained, covering mostly: (i) the lack of a well-articulated medium-term debt management strategy (MTDS); (ii) weak auditing procedures; (iii) lack of procedures to take into account the cost and risks of prospective loans; (iv) inadequate controls of the filing system and; (iv) weak procedures for database access rights. The authorities are continuing efforts to strengthen debt management capacity, notably through training and increased Information Technology resources. They are also planning to request technical assistance from the IMF and the World Bank for the preparation of a MTDS in 2012.

10. **Burkina Faso is classified as a "low debt management capacity" country under the IMF/World Bank Debt Limits Framework**. The authorities are hopeful that further progress on debt management in 2011–12, would support an upgrade to a "higher capacity", leading to some flexibility on borrowing conditions.

V. Authorities Views

11. **The authorities concurred with the DSA results and reaffirmed commitments to prudent debt policies**. They stressed, however, that while maintaining an overall prudent borrowing policy, some flexibility was needed under the current IMF/WB LICs debt limits framework. They believe that access to nonconcessional borrowing would support their efforts to mobilize increased financing and scale-up investment under the new PRSP.

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[3] Although financing from the WAEMU regional bond market is a financing option in the medium to long term, the DSA assumes that the authorities would not contract new domestic debt over the projection period. This assumption is consistent with the fact that so far bonds issuance are done on an annual basis.

V. CONCLUSION

12. **Burkina Faso is subject to a high risk of debt distress.** DSA results indicate that while other debt burden indicators are well-below their policy-dependent indicative thresholds, the PV of debt-to-exports ratio is projected to breach its policy threshold in the long term. In addition, Burkina Faso's external debt is vulnerable to shocks that may adversely affect export earnings. The results highlight the need for continued prudent borrowing policies, fiscal consolidation, and economic diversification.

Table 1. Burkina Faso: External Debt Sustainability Framework, Baseline Scenario, 2008-2031 1/
(In percent of GDP, unless otherwise indicated)

	Actual			Historical Average	0 Standard Deviation	Projections									
	2008	2009	2010			2011	2012	2013	2014	2015	2016	2011-2016 Average	2021	2031	2017-2031 Average
External debt (nominal) 1/	**17.4**	**18.6**	**21.5**			**23.0**	**23.4**	**24.4**	**25.3**	**26.0**	**26.7**		**31.7**	**34.2**	
o/w public and publicly guaranteed (PPG)	17.4	18.6	21.5			23.0	23.4	24.4	25.3	26.0	26.7		31.7	34.2	
Change in external debt	0.6	1.2	2.9			1.5	0.4	1.0	0.9	0.7	0.6		0.9	-0.3	
Identified net debt-creating flows	6.9	3.2	1.9			2.1	5.5	5.5	5.0	5.4	6.0		4.5	1.8	
Non-interest current account deficit	**11.0**	**4.2**	**3.3**	**8.9**	**2.9**	**3.3**	**6.6**	**7.3**	**6.9**	**7.3**	**8.1**		**7.1**	**5.1**	6.7
Deficit in balance of goods and services	16.6	10.9	9.1			9.8	11.4	12.2	11.5	11.7	12.2		10.8	6.8	
Exports	10.0	12.4	18.1			24.1	24.6	24.4	24.5	23.2	21.8		19.0	13.5	
Imports	26.6	23.3	27.2			33.9	36.0	36.6	36.0	34.9	34.0		29.8	20.3	
Net current transfers (negative = inflow)	-5.0	-6.0	-5.3	-4.8	0.8	-6.1	-4.3	-4.5	-4.2	-4.0	-3.8		-3.4	-1.4	-2.6
o/w official	-3.4	-4.4	-3.9			-4.7	-3.1	-3.3	-3.1	-3.0	-2.8		-2.5	-0.6	
Other current account flows (negative = net inflow)	-0.6	-0.7	-0.4			-0.4	-0.4	-0.4	-0.4	-0.4	-0.4		-0.4	-0.3	
Net FDI (negative = inflow)	**-1.3**	**-1.1**	**-0.4**	**-1.1**	**1.4**	**-0.4**	**-0.1**	**-0.7**	**-0.7**	**-0.7**	**-0.7**		**-0.9**	**-1.4**	-1.0
Endogenous debt dynamics 2/	**-2.8**	**0.1**	**-1.0**			**-0.8**	**-1.0**	**-1.1**	**-1.3**	**-1.3**	**-1.4**		**-1.7**	**-1.9**	
Contribution from nominal interest rate	0.2	0.2	0.2			0.2	0.2	0.2	0.3	0.3	0.3		0.4	0.5	
Contribution from real GDP growth	-0.7	-0.5	-1.4			-1.1	-1.2	-1.4	-1.5	-1.6	-1.7		-2.0	-2.3	
Contribution from price and exchange rate changes	-2.3	0.4	0.1			…	…	…	…	…	…		…	…	
Residual (3-4) 3/	**-6.3**	**-2.0**	**0.9**			**-0.5**	**-5.1**	**-4.5**	**-4.1**	**-4.6**	**-5.4**		**-3.6**	**-2.1**	
o/w exceptional financing	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
PV of external debt 4/	…	…	16.4			17.0	17.1	17.5	17.9	18.2	18.4		21.5	23.8	
In percent of exports	…	…	90.2			70.8	69.4	71.8	72.9	78.3	84.4		113.5	175.6	
PV of PPG external debt	**…**	**…**	**16.4**			**17.0**	**17.1**	**17.5**	**17.9**	**18.2**	**18.4**		**21.5**	**23.8**	
In percent of exports	**…**	**…**	**90.2**			**70.8**	**69.4**	**71.8**	**72.9**	**78.3**	**84.4**		**113.5**	**175.6**	
In percent of government revenues	**…**	**…**	**104.8**			**107.8**	**105.5**	**107.7**	**107.7**	**107.0**	**106.0**		**118.0**	**123.2**	
Debt service-to-exports ratio (in percent)	**5.7**	**4.7**	**3.2**			**3.3**	**3.6**	**3.5**	**3.5**	**4.0**	**4.1**		**4.2**	**8.1**	
PPG debt service-to-exports ratio (in percent)	**5.7**	**4.7**	**3.2**			**3.3**	**3.6**	**3.5**	**3.5**	**4.0**	**4.1**		**4.2**	**8.1**	
PPG debt service-to-revenue ratio (in percent)	**4.4**	**4.3**	**3.7**			**5.0**	**5.5**	**5.3**	**5.2**	**5.4**	**5.1**		**4.4**	**5.7**	
Total gross financing need (Billions of U.S. dollars)	0.9	0.3	0.3			0.4	0.8	0.9	0.9	1.0	1.2		1.6	2.7	
Non-interest current account deficit that stabilizes debt ratio	10.4	3.0	0.5			1.7	6.2	6.3	6.0	6.6	7.5		6.2	5.4	
Key macroeconomic assumptions															
Real GDP growth (in percent)	5.2	3.2	7.9	5.7	1.9	5.6	5.8	6.4	6.8	6.8	7.0	6.4	7.2	7.4	7.3
GDP deflator in US dollar terms (change in percent)	15.8	-2.1	-0.7	7.1	7.4	7.2	2.1	1.2	1.2	1.2	1.2	2.3	2.0	2.0	1.8
Effective interest rate (percent) 5/	1.5	1.4	1.3	0.9	0.3	1.2	1.2	1.1	1.1	1.1	1.2	1.2	1.3	1.5	1.3
Growth of exports of G&S (US dollar terms, in percent)	16.0	25.3	56.3	24.1	20.4	50.5	10.5	6.7	8.4	2.2	1.8	13.4	6.3	8.7	5.8
Growth of imports of G&S (US dollar terms, in percent)	31.5	-11.7	25.2	14.8	14.4	40.8	14.9	9.5	6.2	4.8	5.5	13.6	7.0	5.3	5.5
Grant element of new public sector borrowing (in percent)	…	…	…	…	…	41.2	46.9	43.4	42.8	42.3	41.8	43.1	37.9	35.3	37.2
Government revenues (excluding grants, in percent of GDP)	13.1	13.7	15.6			15.8	16.2	16.3	16.6	17.0	17.4		18.3	19.3	18.6
Aid flows (in Billions of US dollars) 7/	115.6	124.4	165.8			0.9	1.0	1.0	1.0	1.1	1.1		1.3	1.4	
o/w Grants	0.3	0.5	0.4			0.7	0.8	0.8	0.8	0.8	0.9		1.0	1.1	
o/w Concessional loans	115.3	123.9	165.4			0.2	0.2	0.2	0.2	0.2	0.3		0.3	0.3	
Grant-equivalent financing (in percent of GDP) 8/	…	…	…			8.9	8.1	7.9	7.6	7.5	7.3		6.1	3.1	5.0
Grant-equivalent financing (in percent of external financing) 8/	…	…	…			76.6	85.5	81.0	80.2	80.0	79.5		71.1	59.6	66.8
Memorandum items:															
Nominal GDP (Billions of US dollars)	8.3	8.4	9.0			10.1	11.0	11.8	12.7	13.8	14.9		22.6	56.2	
Nominal dollar GDP growth	21.8	1.0	7.2			13.1	8.1	7.7	8.1	8.1	8.3	8.9	9.4	9.5	9.2
PV of PPG external debt (in Billions of US dollars)			1.5			1.7	1.9	2.1	2.3	2.5	2.7		4.8	13.3	
(PVt-PVt-1)/GDPt-1 (in percent)						3.0	1.3	1.8	1.8	1.7	1.8	1.9	2.6	2.1	2.4
Gross workers' remittances (Billions of US dollars)	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.1	
PV of PPG external debt (in percent of GDP + remittances)	…	…	16.3			17.0	17.1	17.5	17.9	18.1	18.4		21.5	23.7	
PV of PPG external debt (in percent of exports + remittances)	…	…	89.3			70.3	69.1	71.5	72.6	78.0	84.0		112.4	172.6	
Debt service of PPG external debt (in percent of exports + remittances)	…	…	3.1			3.3	3.6	3.5	3.5	3.9	4.0		4.2	7.9	

Sources: Country authorities; and staff estimates and projections.

1/ Covers public and publicly guaranteed external debt only because private sector debt is unavailable.

2/ Derived as [r - g - ρ(1+g)]/(1+g+ρ+gρ) times previous period debt ratio, with r = nominal interest rate; g = real GDP growth rate, and ρ = growth rate of GDP deflator in U.S. dollar terms.

3/ Includes exceptional financing (i.e., changes in arrears and debt relief); changes in gross foreign assets; and valuation adjustments. For projections also includes contribution from price and exchange rate changes.

4/ Assumes that PV of private sector debt is equivalent to its face value.

5/ Current-year interest payments divided by previous period debt stock.

6/ Historical averages and standard deviations are generally derived over the past 10 years, subject to data availability.

7/ Defined as grants, concessional loans, and debt relief.

8/ Grant-equivalent financing includes grants provided directly to the government and through new borrowing (difference between the face value and the PV of new debt).

Table 2.Burkina Faso: Sensitivity Analysis for Key Indicators of Public and Publicly Guaranteed External Debt, 2011-2031
(In percent)

				Projections				
	2011	2012	2013	2014	2015	2016	**2021**	2031
PV of debt-to GDP ratio								
Baseline	17	17	18	18	18	18	**22**	24
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2011-2031 1/	17	17	18	18	18	18	**23**	30
A2. New public sector loans on less favorable terms in 2011-2031 2	17	18	19	20	22	23	**30**	37
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2012-2013	17	17	18	19	19	19	**22**	25
B2. Export value growth at historical average minus one standard deviation in 2012-2013 3/	17	18	20	20	20	20	**23**	24
B3. US dollar GDP deflator at historical average minus one standard deviation in 2012-2013	17	17	18	19	19	19	**22**	25
B4. Net non-debt creating flows at historical average minus one standard deviation in 2012-2013 4/	17	18	19	19	19	20	**22**	24
B5. Combination of B1-B4 using one-half standard deviation shocks	17	16	15	16	16	16	**20**	23
B6. One-time 30 percent nominal depreciation relative to the baseline in 2012 5/	17	24	25	25	26	26	**30**	34
PV of debt-to-exports ratio								
Baseline	71	69	72	73	78	84	**114**	176
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2011-2031 1/	71	70	72	74	79	84	**119**	224
A2. New public sector loans on less favorable terms in 2011-2031 2	71	73	79	83	93	104	**158**	273
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2012-2013	71	69	72	73	78	84	**113**	175
B2. Export value growth at historical average minus one standard deviation in 2012-2013 3/	71	78	89	90	96	102	**132**	196
B3. US dollar GDP deflator at historical average minus one standard deviation in 2012-2013	71	69	72	73	78	84	**113**	175
B4. Net non-debt creating flows at historical average minus one standard deviation in 2012-2013 4/	71	71	77	78	83	90	**117**	177
B5. Combination of B1-B4 using one-half standard deviation shocks	71	64	58	59	64	69	**97**	156
B6. One-time 30 percent nominal depreciation relative to the baseline in 2012 5/	71	69	72	73	78	84	**113**	175
PV of debt-to-revenue ratio								
Baseline	108	105	108	108	107	106	**118**	123
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2011-2031 1/	108	106	108	109	109	105	**124**	157
A2. New public sector loans on less favorable terms in 2011-2031 2	108	111	118	123	127	131	**164**	191
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2012-2013	108	107	112	112	111	110	**123**	128
B2. Export value growth at historical average minus one standard deviation in 2012-2013 3/	108	111	121	121	119	117	**125**	125
B3. US dollar GDP deflator at historical average minus one standard deviation in 2012-2013	108	108	111	112	111	110	**122**	127
B4. Net non-debt creating flows at historical average minus one standard deviation in 2012-2013 4/	108	108	116	115	114	112	**122**	124
B5. Combination of B1-B4 using one-half standard deviation shocks	108	101	94	95	95	95	**110**	119
B6. One-time 30 percent nominal depreciation relative to the baseline in 2012 5/	108	149	152	152	151	150	**166**	174

Table 2.Burkina Faso: Sensitivity Analysis for Key Indicators of Public and Publicly Guaranteed External Debt, 2011-2031 (continued)
(In percent)

Debt service-to-exports ratio								
Baseline	3	4	4	4	4	4	**4**	8
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2011-2031 1/	3	3	3	3	3	3	**4**	8
A2. New public sector loans on less favorable terms in 2011-2031 2	3	4	3	4	4	5	**6**	14
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2012-2013	3	4	4	4	4	4	**4**	8
B2. Export value growth at historical average minus one standard deviation in 2012-2013 3/	3	4	4	4	5	5	**5**	9
B3. US dollar GDP deflator at historical average minus one standard deviation in 2012-2013	3	4	4	4	4	4	**4**	8
B4. Net non-debt creating flows at historical average minus one standard deviation in 2012-2013 4/	3	4	4	4	4	4	**4**	8
B5. Combination of B1-B4 using one-half standard deviation shocks	3	4	3	3	3	4	**3**	7
B6. One-time 30 percent nominal depreciation relative to the baseline in 2012 5/	3	4	4	4	4	4	**4**	8
Debt service-to-revenue ratio								
Baseline	5	5	5	5	5	5	**4**	6
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2011-2031 1/	5	5	5	5	5	4	**4**	5
A2. New public sector loans on less favorable terms in 2011-2031 2	5	5	5	5	6	6	**6**	10
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2012-2013	5	6	5	5	6	5	**5**	6
B2. Export value growth at historical average minus one standard deviation in 2012-2013 3/	5	5	5	5	6	5	**5**	6
B3. US dollar GDP deflator at historical average minus one standard deviation in 2012-2013	5	6	5	5	6	5	**5**	6
B4. Net non-debt creating flows at historical average minus one standard deviation in 2012-2013 4/	5	5	5	5	6	5	**5**	6
B5. Combination of B1-B4 using one-half standard deviation shocks	5	5	5	5	5	5	**4**	5
B6. One-time 30 percent nominal depreciation relative to the baseline in 2012 5/	5	8	7	7	8	7	**6**	8
Memorandum item:								
Grant element assumed on residual financing (i.e., financing required above baseline) 6/	37	37	37	37	37	37	**37**	37

Sources: Country authorities; and staff estimates and projections.

1/ Variables include real GDP growth, growth of GDP deflator (in U.S. dollar terms), non-interest current account in percent of GDP, and non-debt creating flows.
2/ Assumes that the interest rate on new borrowing is by 2 percentage points higher than in the baseline., while grace and maturity periods are the same as in the baseline.
3/ Exports values are assumed to remain permanently at the lower level, but the current account as a share of GDP is assumed to return to its baseline level after the shock (implicitly assuming an offsetting adjustment in import levels).
4/ Includes official and private transfers and FDI.
5/ Depreciation is defined as percentage decline in dollar/local currency rate, such that it never exceeds 100 percent.
6/ Applies to all stress scenarios except for A2 (less favorable financing) in which the terms on all new financing are as specified in footnote 2.

Table 3.Burkina Faso: Public Sector Debt Sustainability Framework, Baseline Scenario, 2008-2031
(In percent of GDP, unless otherwise indicated)

	Actual					Estimate	Projections								
	2008	2009	2010	Average	Standard Deviation	2011	2012	2013	2014	2015	2016	2011-16 Average	2021	2031	2017-31 Average
Public sector debt 1/	20.1	22.7	24.7			26.5	26.0	26.2	26.4	26.7	27.0		31.7	34.2	
o/w foreign-currency denominated	17.4	18.6	21.5			23.0	23.4	24.4	25.3	26.0	26.7		31.7	34.2	
Change in public sector debt	-0.2	2.7	2.0			1.8	-0.5	0.2	0.2	0.3	0.3		0.9	-0.3	
Identified debt-creating flows	3.0	1.6	4.6			1.1	1.2	0.7	0.3	0.0	-0.2		1.0	-0.4	
Primary deficit	4.1	4.3	5.1	2.2	6.5	3.5	2.5	2.2	1.9	1.7	1.5	2.2	3.3	2.2	2.8
Revenue and grants	17.1	19.6	20.1			22.8	23.1	22.8	22.8	23.1	23.3		22.8	21.3	
of which: grants	4.0	5.9	4.6			7.0	6.9	6.5	6.2	6.1	5.9		4.6	2.0	
Primary (noninterest) expenditure	21.2	23.9	25.3			26.3	25.6	24.9	24.8	24.8	24.8		26.1	23.5	
Automatic debt dynamics	-1.0	-1.9	-0.4			-2.3	-1.3	-1.5	-1.6	-1.7	-1.7		-2.3	-2.5	
Contribution from interest rate/growth differential	-2.1	-0.8	-1.8			-1.3	-1.5	-1.6	-1.8	-1.8	-1.9		-2.3	-2.5	
of which: contribution from average real interest rate	-1.1	-0.2	-0.1			0.0	0.0	-0.1	-0.1	-0.2	-0.2		-0.2	-0.2	
of which: contribution from real GDP growth	-1.0	-0.6	-1.7			-1.3	-1.5	-1.6	-1.7	-1.7	-1.7		-2.1	-2.4	
Contribution from real exchange rate depreciation	1.1	-1.1	1.4			-1.0	0.2	0.2	0.2	0.2	0.2		...	...	
Other identified debt-creating flows	-0.1	-0.8	-0.2			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Privatization receipts (negative)	-0.1	-0.8	-0.2			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Recognition of implicit or contingent liabilities	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Debt relief (HIPC and other)	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Other (specify, e.g. bank recapitalization)	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Residual, including asset changes	-3.2	1.0	-2.6			0.6	-1.7	-0.5	-0.1	0.3	0.5		-0.1	0.0	
Other Sustainability Indicators															
PV of public sector debt	...	...	19.5			20.5	19.7	19.4	19.0	18.8	18.7		21.5	23.8	
o/w foreign-currency denominated	...	...	16.4			17.0	17.1	17.5	17.9	18.2	18.4		21.5	23.8	
o/w external	...	...	16.4			17.0	17.1	17.5	17.9	18.2	18.4		21.5	23.8	
PV of contingent liabilities (not included in public sector debt)	...	...	...			...	...	...	...	...	...		...	...	
Gross financing need 2/	5.0	5.4	6.5			5.0	4.3	3.7	3.4	3.0	2.7		4.1	3.3	
PV of public sector debt-to-revenue and grants ratio (in percent)	…	…	97.0			89.8	85.2	85.0	83.3	81.5	80.3		94.4	111.7	
PV of public sector debt-to-revenue ratio (in percent)	…	…	125.3			129.5	121.4	118.8	114.6	111.1	107.8		118.0	123.2	
o/w external 3/	…	…	104.8			107.8	105.5	107.7	107.7	107.0	106.0		118.0	123.2	
Debt service-to-revenue and grants ratio (in percent) 4/	5.4	5.2	6.7			6.7	7.6	6.9	6.5	5.8	5.4		3.5	5.1	
Debt service-to-revenue ratio (in percent) 4/	7.1	7.4	8.6			9.6	10.8	9.6	8.9	7.9	7.2		4.4	5.7	
Primary deficit that stabilizes the debt-to-GDP ratio	4.3	1.7	3.2			1.7	3.0	1.9	1.7	1.4	1.2		2.4	2.5	
Key macroeconomic and fiscal assumptions															
Real GDP growth (in percent)	5.2	3.2	7.9	5.7	1.9	5.6	5.8	6.4	6.8	6.8	7.0	6.4	7.2	7.4	7.3
Average nominal interest rate on forex debt (in percent)	1.5	1.4	1.3	0.9	0.3	1.2	1.2	1.1	1.1	1.1	1.2	1.2	1.3	1.5	1.3
Average real interest rate on domestic debt (in percent)	-3.5	4.8	2.8	2.9	3.7	5.2	5.9	4.1	4.0	3.9	3.9	4.5	...	...	3.4
Real exchange rate depreciation (in percent, + indicates depreciation)	7.0	-6.7	8.3	-3.2	10.3	-5.0	...	...	...	...	...	...	...	...	...
Inflation rate (GDP deflator, in percent)	7.9	3.4	2.8	3.0	2.5	2.1	2.2	2.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0
Growth of real primary spending (deflated by GDP deflator, in percent)	-0.1	0.2	0.1	0.1	0.1	0.1	0.0	0.0	0.1	0.1	0.1	0.1	0.1	0.1	0.1
Grant element of new external borrowing (in percent)	...	...	...	…	…	41.2	46.9	43.4	42.8	42.3	41.8	43.1	37.9	35.3	...

Sources: Country authorities; and staff estimates and projections.

1/ Public sector debt is defined as gross debt for the general government and nonfinancial public sector.

2/ Gross financing need is defined as the primary deficit plus debt service plus the stock of short-term debt at the end of the last period.

3/ Revenues excluding grants.

4/ Debt service is defined as the sum of interest and amortization of medium and long-term debt.

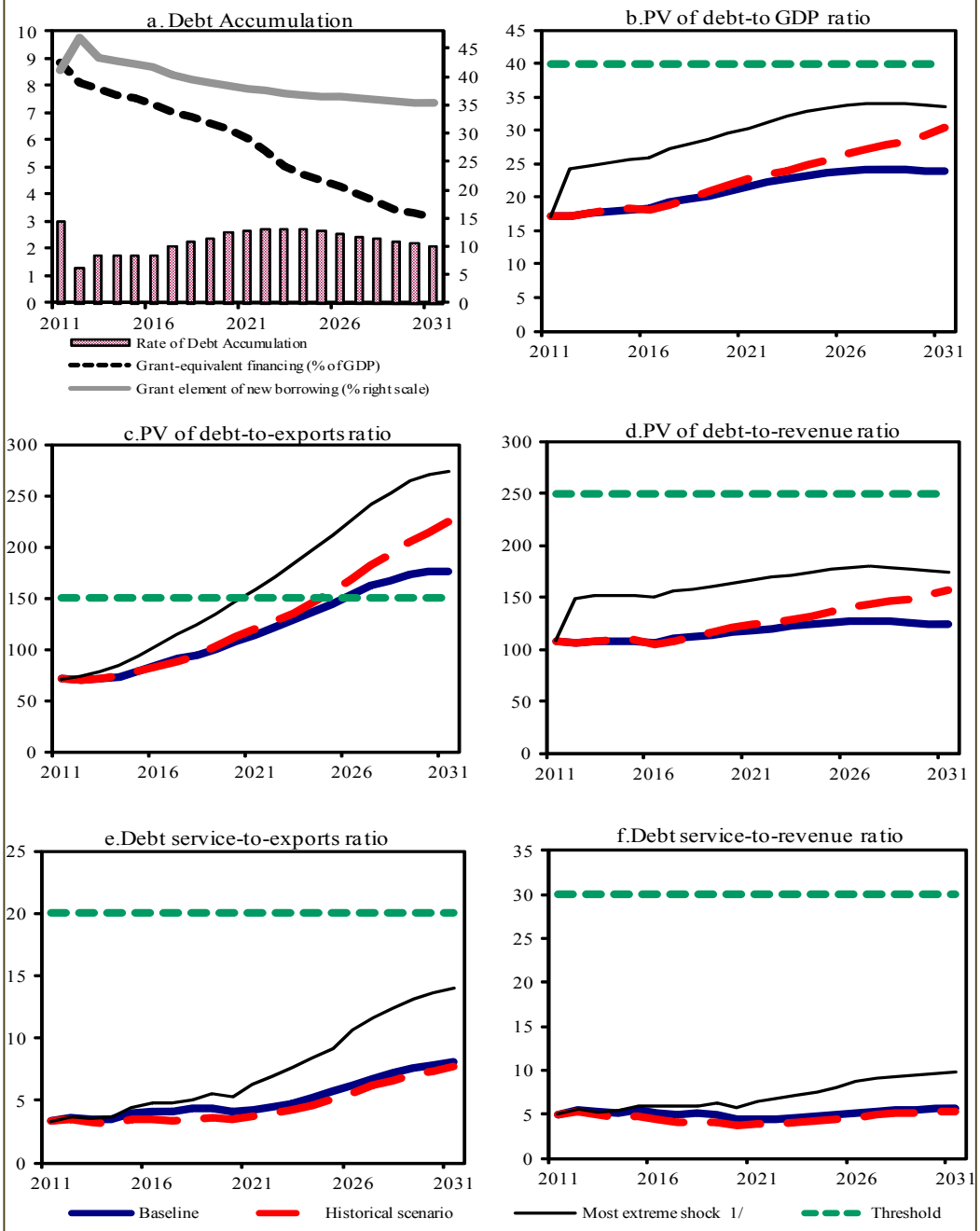
5/ Historical averages and standard deviations are generally derived over the past 10 years, subject to data availability.

Table 4.Burkina Faso: Sensitivity Analysis for Key Indicators of Public Debt 2011-2031

	Projections							
	2011	2012	2013	2014	2015	2016	2021	2031
PV of Debt-to-GDP Ratio								
Baseline	20	20	19	19	19	19	22	24
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	20	19	19	19	20	20	21	22
A2. Primary balance is unchanged from 2011	20	20	20	20	20	21	21	22
A3. Permanently lower GDP growth 1/	20	20	20	20	20	20	25	35
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2012-2013	20	20	21	21	22	22	27	32
B2. Primary balance is at historical average minus one standard deviations in 2012-2013	20	24	29	28	27	27	28	27
B3. Combination of B1-B2 using one half standard deviation shocks	20	22	24	24	24	24	27	30
B4. One-time 30 percent real depreciation in 2012	20	26	25	24	23	22	22	23
B5. 10 percent of GDP increase in other debt-creating flows in 2012	20	27	27	26	25	25	26	26
PV of Debt-to-Revenue Ratio 2/								
Baseline	90	85	85	83	82	80	94	112
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	90	84	84	84	84	85	89	99
A2. Primary balance is unchanged from 2011	90	85	86	86	87	88	93	104
A3. Permanently lower GDP growth 1/	90	86	86	85	85	85	109	161
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2012-2013	90	87	91	92	92	93	117	151
B2. Primary balance is at historical average minus one standard deviations in 2012-2013	90	105	126	123	119	115	122	127
B3. Combination of B1-B2 using one half standard deviation shocks	90	95	106	105	104	103	120	140
B4. One-time 30 percent real depreciation in 2012	90	114	111	105	100	96	98	106
B5. 10 percent of GDP increase in other debt-creating flows in 2012	90	118	117	114	110	107	116	123
Debt Service-to-Revenue Ratio 2/								
Baseline	7	8	7	6	6	5	3	5
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	7	8	7	7	6	6	4	5
A2. Primary balance is unchanged from 2011	7	8	7	7	6	6	4	5
A3. Permanently lower GDP growth 1/	7	8	7	7	6	6	4	7
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2012-2013	7	8	7	7	6	6	4	7
B2. Primary balance is at historical average minus one standard deviations in 2012-2013	7	8	7	8	7	6	4	6
B3. Combination of B1-B2 using one half standard deviation shocks	7	8	7	7	6	6	4	7
B4. One-time 30 percent real depreciation in 2012	7	8	9	8	8	7	5	8
B5. 10 percent of GDP increase in other debt-creating flows in 2012	7	8	8	7	7	6	4	6

Sources: Country authorities; and staff estimates and projections.
1/ Assumes that real GDP growth is at baseline minus one standard deviation divided by the square root of the length of the projection period.
2/ Revenues are defined inclusive of grants.

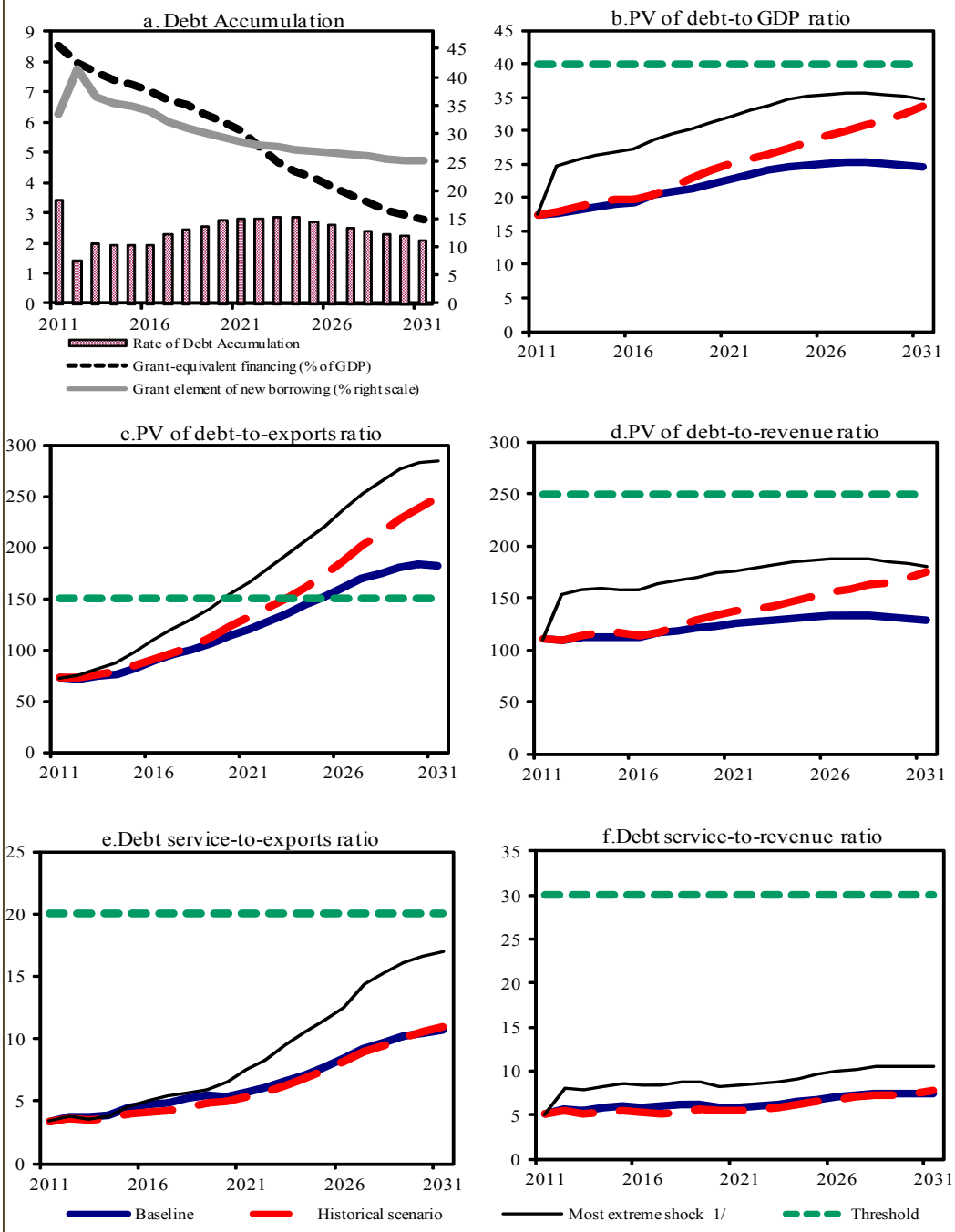
Figure 1. Burkina Faso: Indicators of Public and Publicly Guaranteed External Debt under Alternative Scenarios, 2011-2031 1/



Sources: Country authorities; and staff estimates and projections.

1/ The most extreme stress test is the test that yields the highest ratio in 2021. In figure b. it corresponds to a Previous DSA shock; in c. to a Previous DSA shock; in d. to a Previous DSA shock; in e. to a Previous DSA shock and in figure f. to a Previous DSA shock

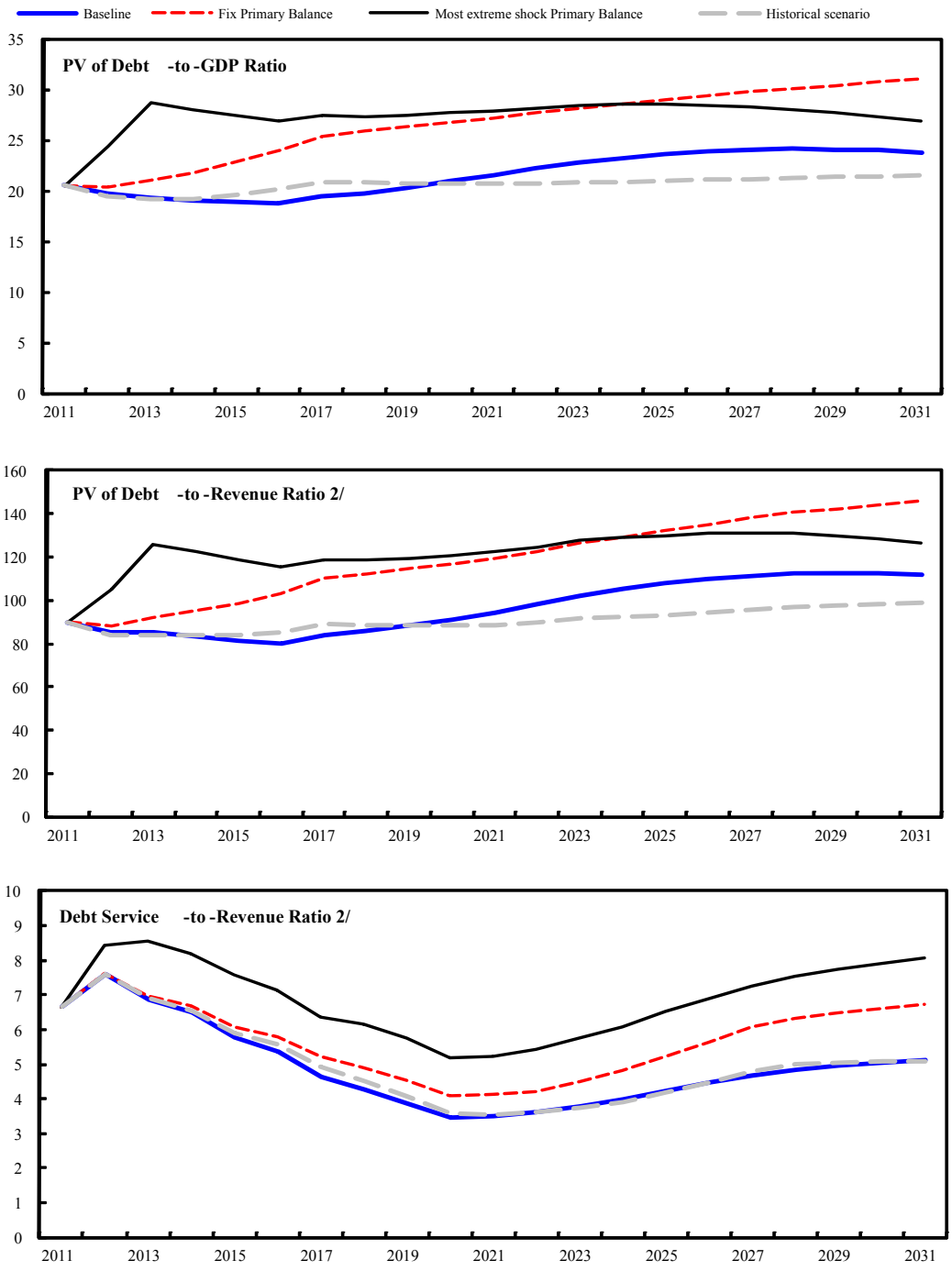
Figure 2. Burkina Faso: Indicators of Public and Publicly Guaranteed External Debt under Country-Specific Alternative Scenarios, 2011-2031 1/



Sources: Country authorities; and staff estimates and projections.

1/ The most extreme stress test is the test that yields the highest ratio in 2021. In figure b. it corresponds to a Previous DSA shock; in c. to a Previous DSA shock; in d. to a Previous DSA shock; in e. to a Previous DSA shock and in figure f. to a Previous DSA shock

Figure 3.Burkina Faso: Indicators of Public Debt Under Alternative Scenarios, 2011-2031 1/



Sources: Country authorities; and staff estimates and projections.
1/ The most extreme stress test is the test that yields the highest ratio in 2021.
2/ Revenues are defined inclusive of grants.